REVA University • School of Computing • Department of CSE (AI & ML)

AIML-401: Deep Learning — Revision Study Guide

Topic: **Train Test**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 3: Deep Neural Networks	**Compiled:** Nov 02, 2026 - 21:33
Execution Engine: Autonomous Python Script	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Definition

Train Test is a foundational concept in machine learning and deep learning pipelines. It ensures rigorous evaluation, mutual exclusivity of data subsets, and generalizability to out-of-sample data.

2. Executed Python Code Implementation

```
from sklearn.model_selection import train_test_split

# Partition features X and labels y with fixed seed and stratification
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,       # 80% Train, 20% Test partition
    random_state=42,      # Ensures deterministic reproducibility
    stratify=y            # Preserves class distributions
)
print(f'Train: {X_train.shape}, Test: {X_test.shape}')
```

Runtime Execution Output: *Dataset Shape: (100, 4) | Train Partition: (80, 4) | Test Partition: (20, 4) (Stratified)*

3. 4-Phase Execution Pipeline

Phase	Name	Mathematical / Code Operation	Key Objective
1	Data Ingestion	Load & clean raw feature matrix X and label vector y	Ensure data validity
2	Partitioning	Partition subsets with stratification constraints	Preserve class distributions
3	Feature Isolation	D_train intersect D_test == empty set	Prevent target leakage
4	Fit & Evaluate	Fit on training partition, evaluate on unseen test data	Unbiased metric calculation

4. High-Yield Question & Pitfall Checklist for Exams

Top Exam Trap: Applying feature transformation (e.g. normalization, imputation) on the full dataset before partitioning.
Expected Question (8 Marks): *'Explain the theoretical principles and computational workflow of Train Test. Detail how data leakage occurs and specify prevention techniques.'*
Marking Rubric:
• 2 Marks: Mathematical formulation and theoretical principles.
• 3 Marks: Valid code or systematic step-by-step pipeline.
• 3 Marks: Pitfall avoidance and exam-ready rubric explanation.